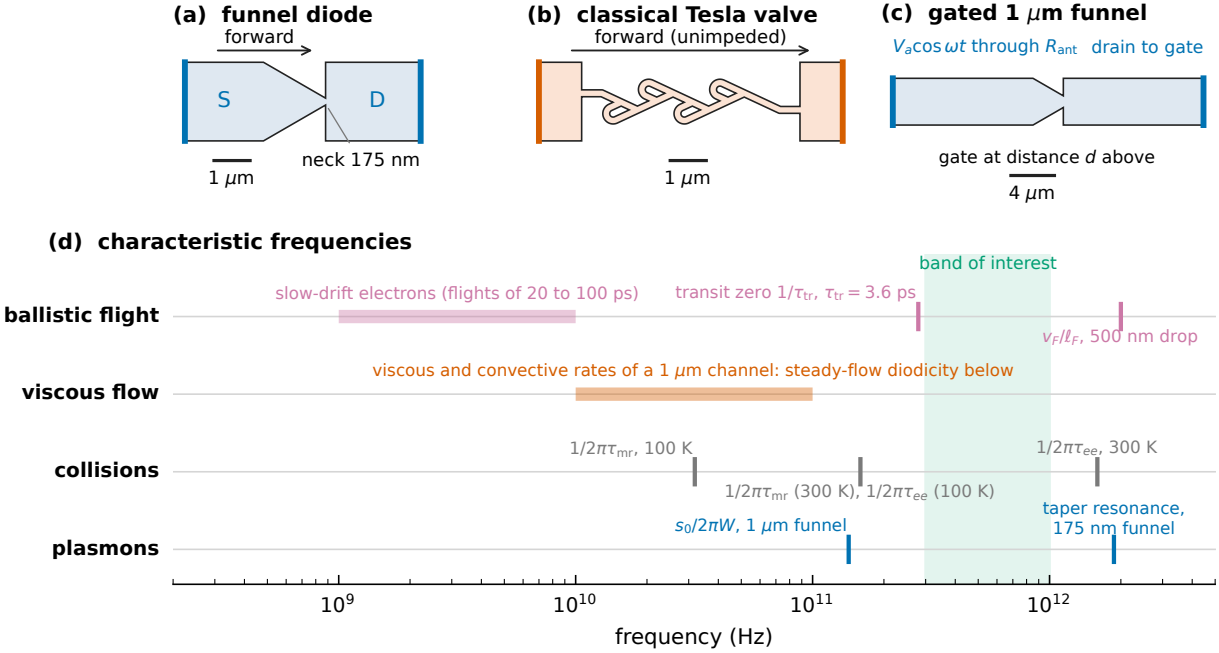
(a) funnel diode
forward
S
D
neck 175 nm
1 μm
(b) classical Tesla valve
forward (unimpeded)
1 μm
(c) gated 1 μm funnel
$V_a \cos \omega t$ through R_{ant}
drain to gate
gate at distance d above
4 μm
(d) characteristic frequencies
band of interest
slow-drift electrons (flights of 20 to 100 ps)
transit zero $1/\tau_{tr}$, $\tau_{tr} = 3.6$ ps
v_F/ℓ_F, 500 nm drop
ballistic flight
viscous and convective rates of a 1 μm channel: steady-flow diodicity below
viscous flow
$1/2\pi\tau_{mr}$, 100 K
$1/2\pi\tau_{ee}$, 300 K
$1/2\pi\tau_{mr}$ (300 K), $1/2\pi\tau_{ee}$ (100 K)
collisions
taper resonance,
175 nm funnel
$s_0/2\pi W$, 1 μm funnel
plasmons
10^9
10^{10}
10^{11}
10^{12}
frequency (Hz)